



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Advanced)

SCORE - II

06-06-2021

JEE(Main + Advance) : Leader & Enthusiast Course (PHASE : SCORE-II)

Test Type : SCORE - II

Test Pattern : JEE (Advanced)

TEST DATE : 06-06-2021

ANSWER KEY

PAPER-2

PHYSICS

SEC-1	Q.	1	2	3	4						
	A.	A,C,D	B	B,C	A,B,D						
SEC-2	Q.	1	2	3	4	5	6				
	A.	B	B	B	C	D	B				
SEC-3	Q.	1	2	3	4	5	6	7	8	9	10
	A.	3	4	8	4	7	8	8	4	6	4

CHEMISTRY

SEC-1	Q.	1	2	3	4						
	A.	A,B,C	A,B,C	A,B,C	B,C						
SEC-2	Q.	1	2	3	4	5	6				
	A.	B	A	D	C	B	B				
SEC-3	Q.	1	2	3	4	5	6	7	8	9	10
	A.	3	7	7	5	8	2	4	7	6	4

MATHEMATICS

SEC-1	Q.	1	2	3	4						
	A.	B	A,C	A,B,C	A,B,C,D						
SEC-2	Q.	1	2	3	4	5	6				
	A.	D	C	B	B	D	D				
SEC-3	Q.	1	2	3	4	5	6	7	8	9	10
	A.	8	5	9	7	8	4	8	1	6	4

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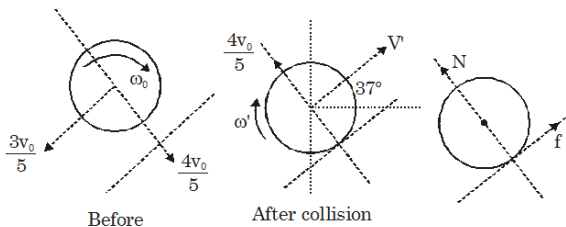
TEST DATE : 06-06-2021

PAPER-2

HINT – SHEET

PHYSICS**SEC-1**

1. Ans (A,C,D)



We can neglect impulse of mg during collision.
 Impulse perpendicular to surface

$$Ndt = 2m \left(\frac{4V_0}{5} \right) \dots (i)$$

Impulse along the surface

$$\mu Ndt = mv' - \left(-m \frac{3v_0}{5} \right)$$

$$\mu Ndt = mv' + \frac{3mv_0}{5} \dots (ii)$$

$$\mu \times 8 \frac{mv_0}{5} = mv' + \frac{3mv_0}{5};$$

$$\frac{8\mu}{5} v_0 = v' + \frac{3v_0}{5} \dots (3)$$

After collision ball will move vertically upwards if

$$v' \cos 37^\circ = \frac{4v_0}{5} \sin 37^\circ; v' = \frac{3}{5} v_0$$

From Eq. (3)

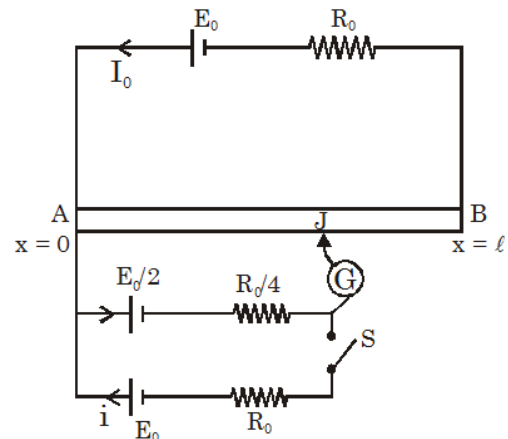
$$\frac{8\mu v_0}{5} = \frac{3}{5} v_0 + \frac{3v_0}{5}; \mu = \frac{6}{8} = \frac{3}{4}$$

Velocity after collision

$$= v' \sin 37^\circ + \frac{4v_0}{5} \cos 37^\circ = v_0 \text{ upwards}$$

$$-\mu Ndt \times R = I\omega' - I\omega_0 \Rightarrow \omega' = \omega_0 - \frac{3v_0}{R}$$

2. Ans (B)



When S is open :

$$R_{AB} = \int_0^l \frac{3R_0 x dx}{\ell^2} = \frac{3R_0}{2}$$

$$I_0 = \frac{E_0}{R_0 + \frac{3R_0}{2}} = \frac{2E_0}{5R_0}$$

$$\frac{E_0}{2} = I_0 \times R_{AJ} = \frac{2E_0}{5R_0} \times \frac{3R_0}{\ell^2} \int_0^x x dx$$

$$x = \ell \sqrt{\frac{5}{6}}$$

When s is closed :

Current in secondary circuit

$$i = \frac{\frac{E_0}{2}}{R_0 + \frac{R_0}{4}} = \frac{2E_0}{5R_0}$$

$$\text{Terminal voltage of battery} = \left[\varepsilon_0 - \frac{2\varepsilon_0}{5} \right] = \frac{3\varepsilon_0}{5}$$

$$\frac{3\varepsilon_0}{5} = I_0 \times R_{AJ} = \frac{2\varepsilon_0}{5R_0} \times \frac{3R_0}{\ell^2} \times \frac{x^2}{2} \Rightarrow x = \ell$$

3. Ans (B,C)

Negative zero error = -1.25

thickness = $18 + 0.34 - (-1.25)$

= 19.59 mm

4. Ans (A,B,D)

Displacement of centre of mass is zero. So take centre of mass as origin then find displacement of all points.

PHYSICS

SEC-2

1. Ans (B)

↑ +ve

$V_{AL} = +2 \text{ m/s}$

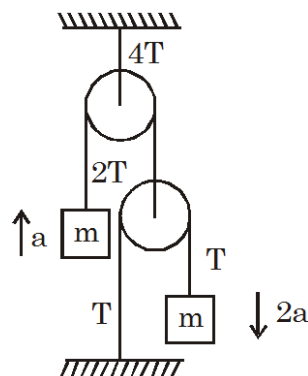
$V_L = +6 \text{ m/s}$

$V_A = V_{AL} + V_L = +8 \text{ m/s}$

$V_{BL} = -2V_{AL} = -4 \text{ m/s}$

$V_B = V_{BL} + V_L = -4 + 6 = +2 \text{ m/s}$

2. Ans (B)



$$mg - T = 2ma \Rightarrow a = \frac{g}{5}$$

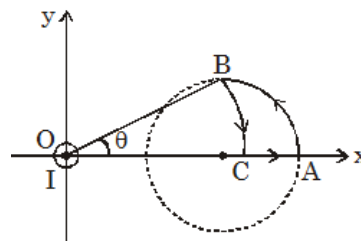
$$2T - mg = ma$$

$$T = mg - \frac{2mg}{5}$$

$$= \frac{3mg}{5}$$

3. Ans (B)

Let segment $OB = OC$ and arc BC is a circular arc with centre at origin. Since the shown closed path $ABCA$ encloses no current, the path integral of magnetic field over this path is zero.



$$\text{Hence } \int_A^B \vec{B} \cdot d\vec{\ell} + \int_B^C \vec{B} \cdot d\vec{\ell} + \int_C^A \vec{B} \cdot d\vec{\ell} = 0$$

Because $\vec{B}$ is perpendicular to segment AC at all

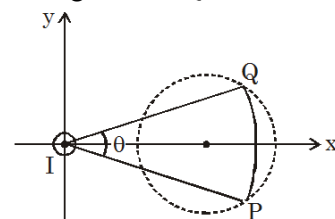
points, therefore $\int_C^A \vec{B} \cdot d\vec{\ell} = 0$

Hence

$$\int_A^B \vec{B} \cdot d\vec{\ell} = \int_B^C \vec{B} \cdot d\vec{\ell} = \frac{\mu_0 I}{2\pi} \frac{OB(\theta)}{OB} = \frac{\mu_0 I}{2\pi} \tan^{-1} \frac{1}{2}$$

4. Ans (C)

Consider two points P and Q lying on dotted circle and equidistant from origin O . We draw a circular arc QP with centre at origin O . The path integral of magnetic field, that is $\int \vec{B} \cdot d\vec{\ell}$, along the dotted circle between two points P and Q is also equal to path integral $\int \vec{B} \cdot d\vec{\ell}$ along the arc QP whose centre is at origin.



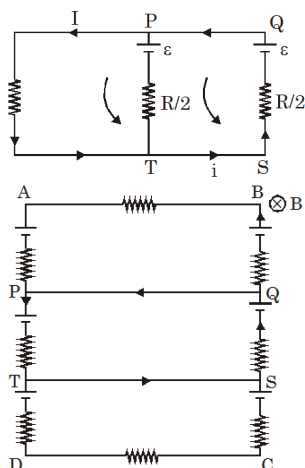
Therefore the path integral of magnetic field $\int \vec{B} \cdot d\vec{\ell}$ along the dotted circle between two points P and Q .

$$= \frac{\mu_0 I}{2\pi} \frac{OP(\theta)}{OP} = \frac{\mu_0 I}{2\pi} \theta$$

The value of θ will be maximum when chord OQ and chord OP will be tangent to the dotted circle, that is, $\theta = \frac{\pi}{3}$.

Hence the required maximum value = $\frac{\mu_0 I}{6}$.

6. Ans (B)



(7) (D) Current in side AB and CD = 0

$$\varepsilon = BV_0 \ell$$

Apply KVL

$$I \times R + (I - i) \times \frac{R}{2} - \varepsilon = 0 \dots (1)$$

$$I \times \frac{R}{2} - \varepsilon + \varepsilon - (I - i) \times \frac{R}{2} = 0 \dots (2)$$

$$\Rightarrow I = \frac{4\varepsilon}{5R}, i = \frac{2\varepsilon}{5R}$$

$$(A) \text{ Current in } R = \frac{4BV_0 I}{5R}$$

(B) Current in AB = 0

(C) Current in CD = 0

$$(D) \text{ Current in } ST = i = \frac{2\varepsilon}{5R} = \frac{2BV_0 \ell}{5R}$$

(8) (D) Magnetic force on wire

Force = force on QS + force on PT

$$= BiI + B(I - i)\ell = \frac{4B^2 \ell^2 V_0}{5R}$$

Let potential at S is zero

$$V_S = 0 \Rightarrow V_C = -\frac{\varepsilon}{2}$$

$$V_Q = \varepsilon - i \times \frac{R}{2} = \frac{4\varepsilon}{5R} \Rightarrow V_B = V_Q + \frac{\varepsilon}{2}$$

$$V_{BC} = V_B - V_C = V_Q + \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$$

$$V_{BC} = \frac{9\varepsilon}{5} = \frac{9BV_0 I}{5}$$

PHYSICS

SEC-3

1. Ans (3)

$$P = \frac{(2)^2 \times 4}{(4 - 2)^3} = 2;$$

$$\frac{\Delta P}{P} = 2 \frac{\Delta x}{x} + \frac{\Delta y}{y} + 3 \frac{\Delta t}{(4 - t)};$$

$$= \frac{2 \times 0.01}{2} + \frac{0.02}{4} + \frac{3 \times 0.01}{2}$$

$$= \frac{\Delta P}{P} = 0.01 + \frac{0.01}{2} + \frac{0.03}{2}$$

$$\Delta P = 0.02 + 0.01 + 0.03$$

$$\Delta P = 0.06$$

$$P \pm \Delta P = 2 \pm 0.06$$

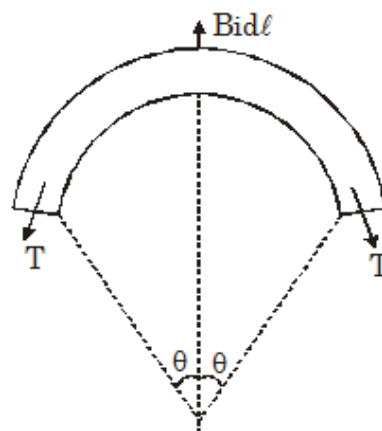
2. Ans (4)

$$2T \sin \theta = Bi\ell$$

$$2T\theta = Bi(R \times 2\theta)$$

$$T = BiR$$

$$v = \sqrt{\frac{T}{\rho S}} = \sqrt{\frac{BiR}{\rho S}}$$



$$B = \frac{v^2 \rho S}{iR} = \frac{10 \times 10 \times 2 \times 10^3 \times 0.2 \times 10^{-4}}{1 \times 1}$$

$$B = 4T$$

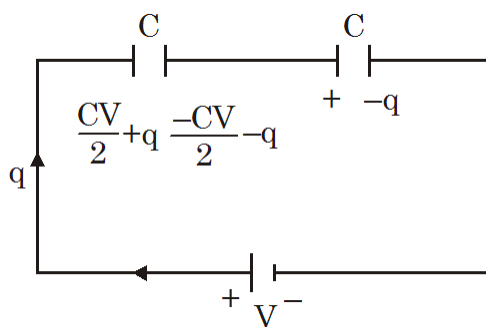
3. Ans (8)

Let charge supplied by battery is q .

Apply KVL

$$\frac{CV}{2} + q + \frac{q}{C} = V$$

$$\frac{CV}{2} + 2q = CV$$



$$2q = \frac{CV}{2}; q = \frac{CV}{4}$$

$$\text{Energy supplied by battery} = q \times V = \frac{CV^2}{4};$$

Energy consumed by capacitors = $U_f - U_i$

$$= \left[\frac{1}{2}C \left(\frac{3V}{4} \right)^2 + \frac{1}{2}C \left(\frac{V}{4} \right)^2 \right] - \left[\frac{1}{2}C \left(\frac{V}{2} \right)^2 + 0 \right]$$

$$= \frac{9CV^2}{32} + \frac{CV^2}{32} - \frac{CV^2}{8} = \frac{3CV^2}{16}$$

Heat loss =

$$\frac{CV^2}{4} - \frac{3CV^2}{16} = \frac{CV^2}{16} = \frac{CV^2}{2 \times 8}$$

4. Ans (4)

The resulted electric field at every point of line

$C_1 C_2$ is zero

$\Rightarrow$ No force on any charge particle

$\Rightarrow$ No change in velocity

5. Ans (7)

$$d_m = \rho_0 r 4\pi r^2 dr$$

$$\text{Inside } g_1 = \frac{Gm}{x^2} = \frac{G \int_0^x \rho_0 \times 4\pi r^2 dr}{x^2}$$

$$g_1 = \frac{G\rho_0 \pi x^4}{x^2}$$

$$g_1 = G\rho_0 \pi x^2$$

Outside-

$$g_2 = \frac{Gm}{y^2} = \frac{G \int_0^R \rho_0 \times 4\pi r^2 dr}{y^2} = \frac{G\rho_0 \pi R^4}{y^2}$$

$$\text{Given } g_1 = g_2 = \frac{1}{2} g_{\text{surface}}$$

$$g_{\text{surface}} = G\rho_0 \pi R^2$$

$$G\rho_0 \pi x^2 = \frac{G\rho_0 \pi R^4}{y^2} = \frac{1}{2} G\rho_0 \pi R^2$$

$$\Rightarrow x = \frac{R}{\sqrt{2}}, y = \sqrt{2}R$$

$$x + y = \frac{R}{\sqrt{2}} + \sqrt{2}R = \frac{3R}{\sqrt{2}}$$

$$\Rightarrow x + y = R\sqrt{\frac{9}{2}}$$

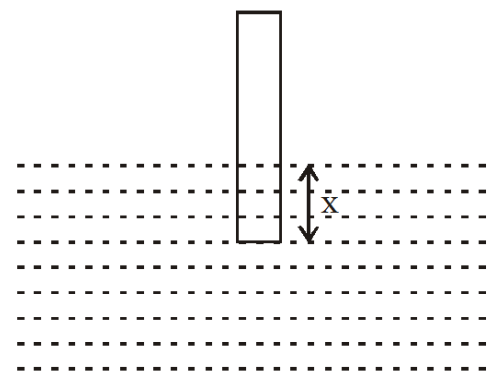
$$\alpha = 9$$

$$\beta = 2$$

6. Ans (8)

$$A\ell\sigma g - A x \rho g = A\ell\sigma a$$

$$a = g - \frac{\rho g x}{\sigma \ell}$$



$$\int_0^v v dv = \int_0^x \left(g - \frac{\rho g x}{\sigma \ell} \right) dx$$

$$\frac{V^2}{2} = gx - \frac{\rho g x^2}{2\sigma \ell}$$

At maximum displacement

$$v = 0$$

$$x = \frac{2\sigma \ell}{\rho} = 8m$$

7. Ans (8)

By the conservation of energy

$$\frac{1}{2}mV_0^2 = mgh + \frac{1}{2}mv^2 \dots (1)$$

By the conservation of angular momentum

$$\frac{mV_0R}{2} = mVR \dots (2)$$

From (1) & (2)

$$V_0 = \sqrt{\frac{8gh}{3}}$$

$$K = 8$$

8. Ans (4)

$$v = \frac{C^2}{\sqrt{t}} \Rightarrow v^2 t = C^4$$

Differentiate w.r.t. 't'

$$v^2 \times 1 + t \times 2v \frac{dv}{dt} = 0$$

$$\Rightarrow \frac{dv}{dt} = -\frac{v}{2t} \dots (1)$$

$$\frac{1}{\text{slope}} = \frac{dv^2}{dt} = 2v \frac{dv}{dt}$$

$$\Rightarrow \frac{dv}{dt} = \frac{1}{\text{slope} \times 2v} \dots (2)$$

$$\text{From (1) and (2)} -\frac{v}{2t} = \frac{1}{\text{slope} \times 2v}$$

$$-\frac{4}{2t} = \frac{1}{-1 \times 2 \times 4}$$

$$\Rightarrow t = 16$$

$$C^2 = v\sqrt{t} = 4\sqrt{16} = 4 \times 4$$

$$\Rightarrow C = 4$$

9. Ans (6)

$$N_1 = 2N_2$$

$$N_0 e^{-\lambda_1 t} = 2N_0 e^{-\lambda_2 t}$$

$$\Rightarrow \frac{1}{t} = \frac{1}{T_2} - \frac{1}{T_1}$$

10. Ans (4)

$$\frac{\Delta Q}{\Delta t} = K_A A \frac{dT}{dx}$$

$$\frac{dT}{dx} \text{ in conductor A} = \text{slope of graph} = \frac{4}{3}$$

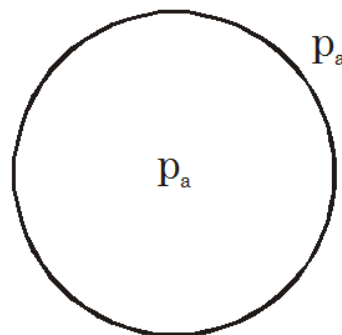
$$\frac{dT}{dx} \text{ in conductor B} = \text{slope of graph} = 1$$

Since both conductors are connected in series, same heat current will flow in A and B.

$$120 \times \frac{4}{3} = 160$$

11. Ans (8)

Inside pressure must be $\frac{4T}{r}$ greater than outside pressure in bubble. This excess pressure is provided by charge on bubble.



$$\frac{4T}{r} = \frac{\sigma^2}{2\epsilon_0}$$

$$\frac{4T}{r} = \frac{Q^2}{16\pi^2 r^4 \times 2\epsilon_0} \dots \dots \left[\sigma = \frac{Q}{4\pi r^2} \right]$$

$$Q = 8\pi r \sqrt{2rT\epsilon_0}$$

12. Ans (4)

$$t = \frac{T}{4} = \frac{2\pi}{4} \sqrt{\frac{I}{pE}}$$

$$I = \frac{4t^2}{\pi^2} pE$$

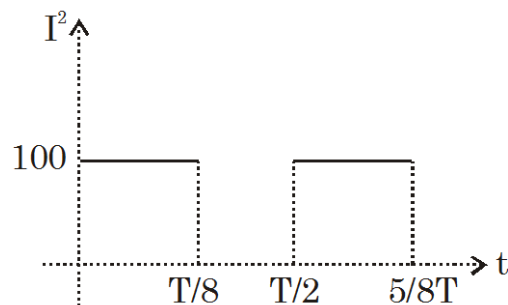
13. Ans (3)

CA is adiabatic, BC is isothermal as per slope.

$$\text{Hence } V_1 = \frac{P_0 V_0}{3P_0} = \frac{V_0}{3}$$

$$\text{Using } PV^\gamma = \text{const} \Rightarrow \gamma = \frac{\ln 6}{\ln 3}$$

14. Ans (5)



$$I_{\text{rms}} = \left[\frac{\int I^2 dt}{\int dt} \right]^{\frac{1}{2}}$$

$$= \left[\frac{100 \frac{T}{8} + 100 \frac{T}{8}}{T} \right]^{\frac{1}{2}}$$

$$= 5A$$

CHEMISTRY

SEC-1

2. **Ans (A,B,C)**

Phase (A) and Phase (B) are at equilibrium, thus

$$\Delta G_{P.T} = G_B - G_A = 0.$$

$$\text{So, } G_A = G_B$$

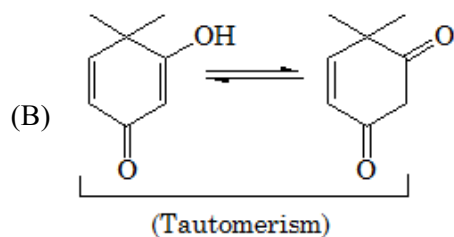
4. **Ans (B,C)**

(A) Incorrect

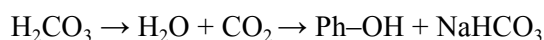
In abbreviated form of dipeptides, amino acid having free amino

group is written first so

Abb. Form : ALA-GLY not GLY-ALA



(C) Correct

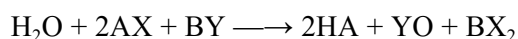


→ No reaction.

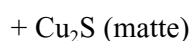
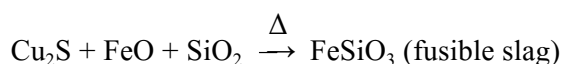
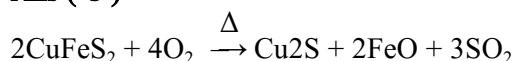
CHEMISTRY

SEC-2

2. **Ans (A)**

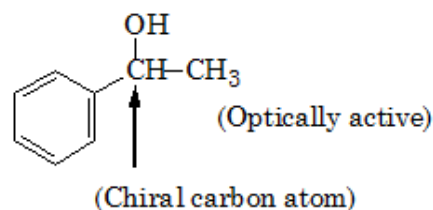


4. **Ans (C)**



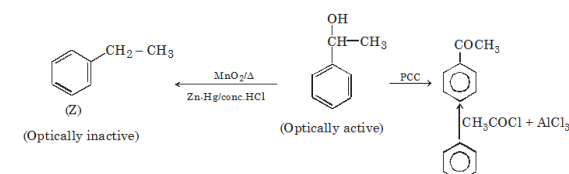
5. **Ans (B)**

The compound (X)



Observations (i), (ii), (iii), (iv), (vi) and (vii) support it.

6. **Ans (B)**



Hence choice (B) is correct while (A), (C) & (D) are incorrect.

CHEMISTRY

SEC-3

1. **Ans (3)**

d,f,c

2. **Ans (7)**

(vi) $\text{O}_2\text{F}_2 < \text{H}_2\text{O}_2$ Bond length of O-O bond

3. **Ans (7)**

$$PT = \text{constant} \setminus PV^{1/2} = \text{constant}$$

$$\therefore C = C_v + R/1-n = 3R/2 + 2R = 7R/2$$

$$\therefore q = nCDT = 2 \times 7R/2 (300 - 600) = -2100R$$

4. **Ans (5)**

$$\frac{112}{22400} \times 6 \times 10^{23} \times 16 \times 10^{-22} = 4.8 \text{ m}^2$$

5. **Ans (8)**

$$x = 6$$

$$y = 1$$

$$z = 3$$

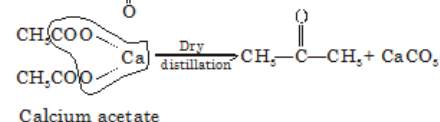
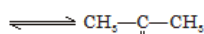
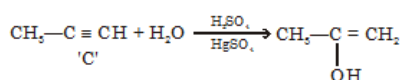
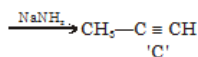
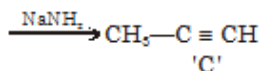
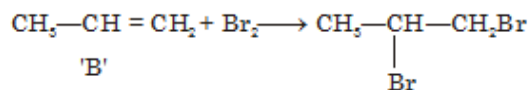
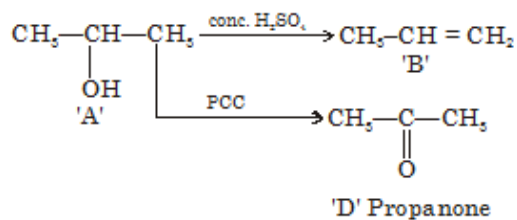
6. **Ans (2)**

$$2 + 6 + 3n - 12 - 2 = 0 \quad [\text{Si}_4\text{O}_{11}]_n^{6n-}$$

$$3n - 6 = 0$$

$$n = 2$$

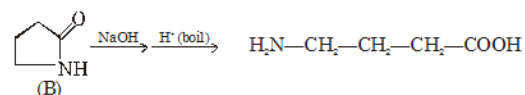
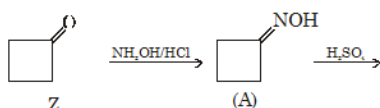
7. Ans (4)



The molecular weight of 'C' (C_3H_4) is $36 + 4 = 40 \text{ g mol}^{-1}$

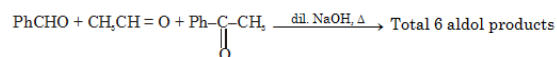
$= 4 \times 10$ hence $Y = 4$

8. **Ans (7)**

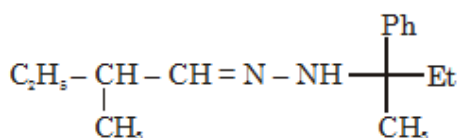


Molar mass of compound Z is 70. So $x = 7$

9. **Ans (6)**



10. **Ans (4)**

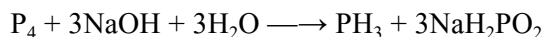


- (I) $(-, +)$ (Z)
 (II) $(+, +)$ (Z)
 (III) $(-, +)$ E
 (IV) $(+, +)$ E

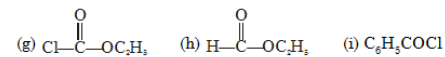
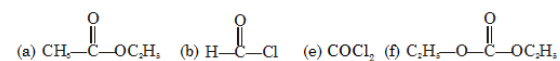
11. Ans (7)

Formula will be = X_3Y_4

12. Ans (3)

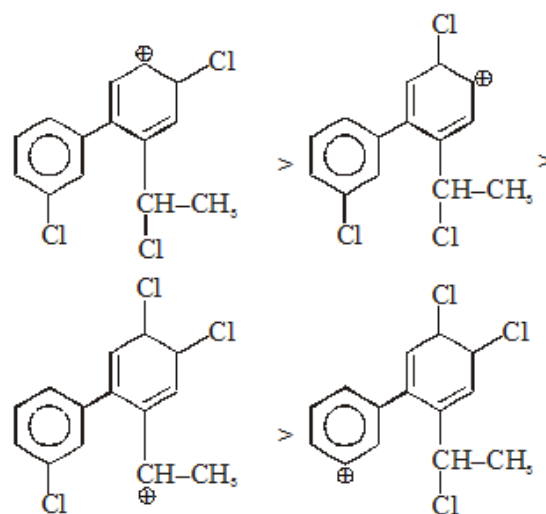


13. Ans (7)



14. **Ans (4)**

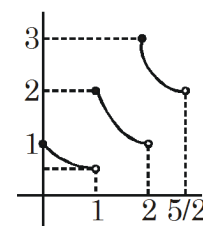
Rate of SN_1 reaction $\propto$ Stability of carbocation
Stability of carbocation



MATHEMATICS

SEC-1

1. **Ans (A,B)**



$$f(x) = \begin{cases} \frac{1}{x+1} & , \quad 0 \leq x < 1 \\ \frac{2}{x} & , \quad 1 \leq x < 2 \\ \frac{3}{x-1} & , \quad 2 \leq x < 5/2 \end{cases}$$

Clearly $f(x)$ is discontinuous bijective function

2. Ans (A,C)

Let $A' = (x_2, y_2, z_2)$ be image of $(2, 1, 6)$ about

mirror $x + y - 2z = 3$, then

$$\frac{x_2 - 2}{1} = \frac{y_2 - 1}{1} = \frac{z_2 - 6}{-2}$$

$$= -2 \left[\frac{2 + 1 - 12 - 3}{1^2 + 1^2 + 2^2} \right] = 4$$

$$\Rightarrow A' \equiv (6, 5, -2)$$

$$\frac{x - 2}{3} = \frac{y - 1}{4} = \frac{z - 6}{5} = \lambda$$

$x = 2 + 3\lambda, y = 4\lambda + 1, z = 6 + 5\lambda$ lies on plane

than B is point where line cuts plane then B $\propto (-$

$$10, -15, -14)$$

The equation of reflected ray will be

$$\frac{x + 10}{6 + 10} = \frac{y + 15}{5 + 15} = \frac{z + 14}{-2 + 14}$$

$$\Rightarrow \frac{x + 10}{16} = \frac{y + 15}{20} = \frac{z + 14}{12}$$

$$\Rightarrow \frac{x + 10}{4} = \frac{y + 15}{5} = \frac{z + 14}{3}$$

3. Ans (A,B,C)

$$\text{Given } \vec{a} \times \vec{b} + \vec{b} \times \vec{c} = p\vec{a} + q\vec{b} + r\vec{c} \quad \dots(1)$$

Take dot product with $\vec{a}$

$$[\vec{a} \ \vec{b} \ \vec{c}] = p + q \cos \theta + r \cos \theta \quad \dots(2)$$

$$\text{Since } \vec{a} \cdot \vec{a} = |\vec{a}|^2 = 1, \vec{a} \cdot \vec{b} = \cos \theta$$

$$\text{and } \vec{a} \cdot \vec{c} = \cos \theta$$

taking $(\cdot)$ product with $\vec{b}$ & $\vec{c}$

$$0 = p \cos \theta + q + r \cos \theta \quad \dots(3)$$

$$[\vec{a} \ \vec{b} \ \vec{c}] = p \cos \theta + q \cos \theta + r \quad \dots(4)$$

from (2) and (4) we get

$$p = r$$

Adding (2), (3) and (4)

$$2 [\vec{a} \ \vec{b} \ \vec{c}] = (2 \cos \theta + 1)(p + q + r)$$

$$\Rightarrow p + q + r = \frac{2 [\vec{a} \ \vec{b} \ \vec{c}]}{2 \cos \theta + 1} \quad \dots(5)$$

multiplying equation (5) by $\cos \theta$ and subtracting it from (2)

$$[\vec{a} \ \vec{b} \ \vec{c}] - \frac{2 [\vec{a} \ \vec{b} \ \vec{c}] \cos \theta}{2 \cos \theta + 1} = p(1 - \cos \theta)$$

$$p = \frac{[\vec{a} \ \vec{b} \ \vec{c}]}{(2 \cos \theta + 1)(1 - \cos \theta)}$$

$$\text{similarly } q = \frac{-2 [\vec{a} \ \vec{b} \ \vec{c}]}{(2 \cos \theta + 1)(1 - \cos \theta)}$$

$$\text{Now } [\vec{a} \ \vec{b} \ \vec{c}]^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & \cos \theta & \cos \theta \\ \cos \theta & 1 & \cos \theta \\ \cos \theta & \cos \theta & 1 \end{vmatrix}$$

$$= (1 + 2 \cos \theta)(1 - \cos \theta)^2$$

$$[\vec{a} \ \vec{b} \ \vec{c}] = \sqrt{1 + 2 \cos \theta} (1 - \cos \theta)$$

Since θ is acute $\cos \theta > 0$

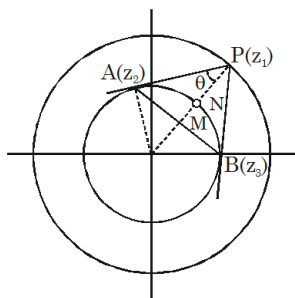
$$\text{Hence } p = r = \frac{1}{\sqrt{1 + 2 \cos \theta}},$$

$$q = \frac{-2 \cos \theta}{\sqrt{1 + 2 \cos \theta}}$$

MATHEMATICS

SEC-2

4. Ans (A, B, C, D)



$$OA = OB = 1; OP = 2$$

$$\sin \angle OPA = \frac{1}{2} \quad \angle OPA = \frac{\pi}{6} \quad \angle APB = \frac{\pi}{3}$$

OM $\perp$ AB so $\triangle APB$ is equilateral, so (D) is true.

$$OM = \frac{1}{2}; MN = \frac{1}{2} \quad PN = 1 \text{ so}$$

centroid lies on $|z| = 1$... (A)

$$\left| \frac{z_1 + z_2 + z_3}{3} \right| = 1$$

$$|z_1 + z_2 + z_3|^2 = 9$$

$$(z_1 + z_2 + z_3)(\bar{z}_1 + \bar{z}_2 + \bar{z}_3) = 9$$

$$\text{or } \left(\frac{4}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} \right) \left(\frac{4}{\bar{z}_1} + \frac{1}{\bar{z}_2} + \frac{1}{\bar{z}_3} \right) = 9$$

... (C)

$$\angle AOB = \pi - \frac{\pi}{3} = \frac{2\pi}{3} \quad \dots (B)$$

1. Ans (D)

$$(A) P\left(\frac{Y}{X_5}\right) = P\left(\frac{Y}{X_6}\right) = \frac{P(Y \cap X_6)}{P(X_6)}$$

$$P(Y \cap X_6) = P(AB \ AB \ AB) + P(AB \ BA \ AA) + P(BA \ AB \ AA) + P(BA \ BA \ AA)$$

$$= 4 \cdot \frac{4}{81} \cdot \frac{4}{9} = \frac{64}{729}$$

$$P(X_6) = P(ABBA) + P(ABAB) + P(BABA) +$$

$$P(BAAB) = 4 \cdot \frac{4}{81}$$

$$\Rightarrow P\left(\frac{Y \cap X_6}{X_6}\right) = \frac{4}{9}$$

$$(B) P\left(\frac{Z}{X_4}\right) = \frac{P(Z \cap X_4)}{P(X_4)}$$

$$P(Z \cap X_4) = P(AB \ BB) + P(BA \ BB) + P(AB \ AB \ BB) + P(AB \ BA \ BB) + P(BA \ AB \ BB) + P(BA \ BA \ BB)$$

$$= \frac{4}{81} + 4 \cdot \frac{4}{81} \cdot \frac{1}{9} = \frac{52}{729}$$

$$P(X_4) = 1 - P(AA) - P(BB)$$

$$= 1 - \left(\frac{4}{9} + \frac{1}{9} \right) = \frac{4}{9}$$

$$P\left(\frac{Z}{X_4}\right) = \frac{52}{729} \times \frac{9}{4} = \frac{13}{81}$$

(C) Match will end at 2, 4 or 6 sets only so

$$P(X_{2k-1}) = P(X_{2k}) \quad \forall k \in \{1, 2, 3\}$$

(D) Hence D is incorrect

$$\rightarrow P\left(\frac{Z}{X_1}\right) = P(Z) = \frac{133}{729}$$

2. Ans (C)

$$P\left(\frac{Y}{W}\right) = \frac{P(Y \cap W)}{P(W)}$$

Where W is the event that A has won the third set

$$P(Y \cap W) = P(AB \ AA) + P(BA \ AA) + P(AB \ AB \ AA) + P(BA \ AB \ AA)$$

$$= 2 \cdot \frac{2}{9} \left(\frac{4}{9} \right) + 2 \cdot \left(\frac{4}{81} \right) \left(\frac{4}{9} \right)$$

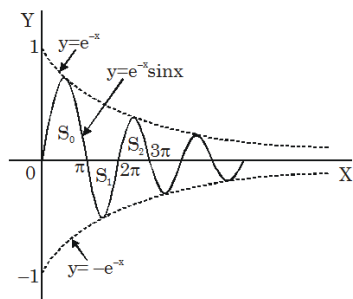
$$= \frac{6}{81} + \frac{32}{729} = \frac{144 + 32}{729} = \frac{176}{729}$$

$$P(W) = P(ABA) + P(BAA)$$

$$= 2 \cdot \frac{2}{9} \cdot \frac{2}{3} = \frac{8}{27}$$

$$P\left(\frac{Y}{W}\right) = \frac{P(Y \cap W)}{P(W)} = \frac{\frac{176}{729}}{\frac{8}{27}} = \frac{176}{27} \cdot \frac{27}{8} = \frac{22}{27}$$

3. Ans (B)



$-1 \leq \sin x \leq 1$ the curve $y = e^{-x} \sin x$ is bounded by $y = e^{-x}$ and $y = -e^{-x}$

Now S_n ; $n\pi \leq x \leq (n+1)\pi$

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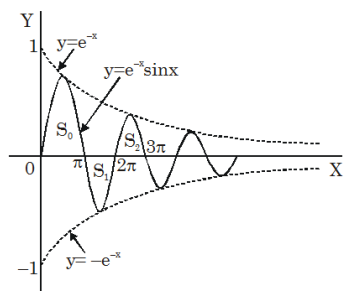
$$S_n = (-1)^n \int_{n\pi}^{(n+1)\pi} e^{-x} \sin x \, dx$$

$$= \frac{e^{-n\pi}}{2} (1 + e^{-\pi})$$

$$S_0 = \frac{1}{2} (1 + e^{-\pi}) \text{ and } \frac{S_{n+1}}{S_n} = e^{-\pi}$$

$$\sum_{n=0}^{\infty} S_n = \frac{\frac{1}{2} (1 + e^{-\pi})}{1 - e^{-\pi}} \quad \left[S_{\infty} = \frac{a}{1-r} \right]$$

4. Ans (B)



$-1 \leq \sin x \leq 1$ the curve $y = e^{-x} \sin x$ is bounded by $y = e^{-x}$ and $y = -e^{-x}$

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$$= \frac{e^{-n\pi}}{2} (1 + e^{-\pi})$$

$$S_0 = \frac{1}{2} (1 + e^{-\pi}) \text{ and } \frac{S_{n+1}}{S_n} = e^{-\pi}$$

$$\sum_{n=0}^{\infty} S_n = \frac{\frac{1}{2} (1 + e^{-\pi})}{1 - e^{-\pi}} \quad \left[S_{\infty} = \frac{a}{1-r} \right]$$

5. Ans (D)

$$D_n = \begin{vmatrix} a & 1 & 0 & 0 & 0 & \dots & 0 \\ 1 & a & 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & a & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & a & \dots & \dots & 0 \\ 0 & 0 & 0 & 1 & a & \dots & 0 \\ 0 & 0 & 0 & 0 & 1 & \dots & a \end{vmatrix}$$

expanding it from 1st Row

$$D_n = aD_{n-1} - \begin{vmatrix} 1 & 1 & 0 & 0 & 0 & \dots \\ 0 & a & 1 & 0 & 0 & \dots \\ 0 & 1 & a & 1 & 0 & \dots \\ \vdots & 0 & 1 & a & \dots & \dots \\ 0 & \dots & \dots & \dots & \dots & \dots \end{vmatrix}$$

Applying $C_2 \rightarrow C_2 - C_1$

$$D_n = aD_{n-1} - \begin{vmatrix} 1 & 0 & 0 & 0 & 0 & \dots & 0 \\ 0 & a & 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & a & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & a & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \dots & \vdots \\ 0 & 0 & 0 & \dots & \dots & \dots & 0 \end{vmatrix}$$

$$D_n = aD_{n-1} - D_{n-2}$$

$$a = 2 \quad D_n = 2D_{n-1} - D_{n-2}$$

$$D_n + D_{n-2} = 2D_{n-1}$$

i.e., D_{n-2}, D_{n-1}, D_n are in A.P.

$$D_1 = 2 \quad D_2 = \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = 3$$

$$D_3 = 4 \dots D_{2017} = 2018$$

6. **Ans (D)**

$$a = 1$$

$$D_n = D_{n-1} - D_{n-2}$$

$$D_1 = 1 \quad D_2 = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 0$$

$$D_3 = D_2 - D_1 = -1$$

$$D_4 = -1$$

$$D_5 = 0$$

$$D_6 = 1$$

$$D_7 = 1$$

$$D_8 = 0$$

$$D_9 = -1$$

$$D_{10} = -1$$

$$D_{11} = 0$$

$$D_{12} = 1$$

$$D_{13} = 1$$

$$\text{i.e., } D_1 = D_6 = D_7 = D_{12} = D_{13} = 1$$

$$D_2 = D_5 = D_8 = D_{11} = 0$$

$$D_3 = D_4 = D_9 = D_{10} = -1$$

$$6\lambda, 6\lambda + 1 \text{ type no. gives } 1$$

$$6\lambda + 2, 6\lambda + 5 \text{ gives } 0$$

$$6\lambda + 3, 6\lambda + 4 \text{ gives } -1$$

$$\text{Hence } \sum_{k=1}^{2017} |D_k| = \frac{4}{6}(2016) + 1 \\ = 2(672) + 1 = 1344 + 1 = 1345$$

MATHEMATICS

SEC-3

1. **Ans (8)**

$$\Delta = 0 \text{ \& } a_0 + a_1 + a_2 \neq 0$$

$$\text{then } a_0 = a_1 = a_2$$

$$\Delta = -[(a_0^3 + a_1^3 + a_2^3) - 3a_0a_1a_2]$$

$$\text{putting } x = 0$$

$$a_0 = 1$$

$$\text{coefficient of } x = 4a$$

$$\text{coefficient of } x^2 = 4b + 6a^2$$

$$1 = 4a = 6a^2 + 4b$$

$$\Rightarrow a = \frac{1}{4}; b = \frac{5}{32} \quad 5 \times \left(\frac{1/4}{5/32}\right) = 8$$

2. **Ans (5)**

$$z_1 = (8\sin\theta + 7\cos\theta) + i(\sin\theta + 4\cos\theta)$$

$$z_2 = (\sin\theta + 4\cos\theta) + i(8\sin\theta + 7\cos\theta)$$

$$z_1 = a + ib \quad z_2 = b + ia$$

$$z_1 z_2 = ab - ab + i(a_2 + b_2) = x + iy$$

$$\Rightarrow x = 0; y = a_2 + b_2$$

$$\text{Now } x + y = a_2 + b_2$$

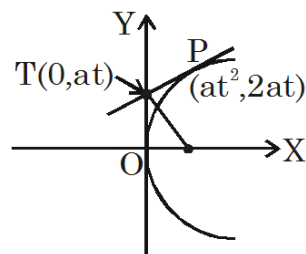
$$M = (x + y)_{\max}$$

$$\Rightarrow x + y = (8\sin\theta + 7\cos\theta)^2 + (\sin\theta + 4\cos\theta)^2 \\ = 65 + 60\sin 2\theta$$

$$(x + y)_{\max} = 125$$

$$m = 125 \Rightarrow M^{1/3} = 5$$

3. **Ans (9)**



(h, k) is mid point of PT then

$$h = \frac{at^2}{2}; k = \frac{2at + at}{2}$$

$$\Rightarrow 2h = a \cdot \frac{4k^2}{9a^2}$$

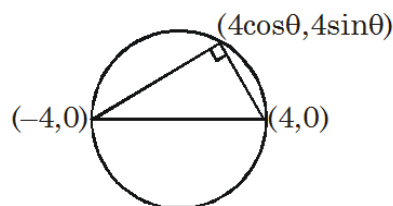
$$\Rightarrow 2y^2 = 9ax$$

$$a = 2$$

$$y^2 = 9x$$

$$L.R = 9$$

4. **Ans (7)**



$$\text{Area} = \frac{1}{2} \times 8 \times 4\sin\theta = |16\sin\theta|$$

If area is integer then $\sin\theta$ can be equal to

$$\frac{1}{16}, \frac{2}{16}, \frac{3}{16}, \dots, \frac{15}{16}$$

15 in all four quadrants and 2 are on axes

$$\left[\frac{62}{8}\right] = 7$$

5. Ans (8)

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x) + f(h) + 2xh(x+h) - \frac{1}{3} - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} + 2x^2 = f'(0) + 2x^2 \\ \text{also} \\ \lim_{h \rightarrow 0} \frac{3f(h) - 1}{6h} &= \lim_{h \rightarrow 0} \frac{f(h) - \frac{1}{3}}{2h} = \frac{f'(0)}{2} = \frac{2}{3} \end{aligned}$$

$$\text{so } f'(x) = \frac{4}{3} + 2x^2$$

$$f(x) = \frac{2x^3}{3} + \frac{4}{3}x + \lambda \text{ or } f(0) = \lambda = \frac{1}{3}$$

$$f(2) = \frac{25}{3} \quad [f(2)] = 8$$

6. Ans (4)

$$f(x) = f(6-x) \quad \dots(i)$$

$$f'(x) = -f'(6-x) \quad \dots(ii)$$

Putting $x = 0, 2, 5$

$$f'(0) = f'(6) = f'(2) = f'(4) = f'(5) = f'(1) = 0$$

and since $f'(3) = -f'(3)$

$$\Rightarrow f'(3) = 0$$

So $f'(x) = 0$ has minimum seven (7) roots in $[0, 6]$

Now if $h(x) = f'(x)f''(x)$

$$h'(x) = (f''(x))^2 + f'(x)f'''(x)$$

$h(x) = 0$ has minimum $7 + 6 = 13$ roots so $h'(x)$ will have minimum 12 zeros is $[0, 6]$

$$\text{So } \frac{12}{3} = 4$$

7. Ans (8)

Eliminating t will give

$$y^2(x-1) = 1$$

equation of tangent at $(2, 1)$ is $x + 2y = 4$

solving with curve,

$$\text{we get } x = 5 \text{ and } y = -\frac{1}{2}$$

$$Q = \left(5, -\frac{1}{2}\right) \text{ or } |PQ| = \frac{3\sqrt{5}}{2}$$

8. Ans (1)

$$f(x) + f(y) = f\left(\frac{x+y}{1-xy}\right) \quad \dots(i)$$

Putting $x = y = 0$, we get $f(0) = 0$

Putting $y = -x$, we get

$$f(x) + f(-x) = f(0)$$

$$\text{or, } f(-x) = -f(x) \quad \dots(ii)$$

$$\text{Also, } \lim_{x \rightarrow 0} \frac{f(x)}{x} = 2$$

$$\text{Now, } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad \dots(iii)$$

$$= \lim_{h \rightarrow 0} \frac{f(x+h) + f(-x)}{h} \quad [\text{Using (ii)}]$$

$$= \lim_{h \rightarrow 0} \frac{f\left(\frac{x+h-x}{1-(x+h)(-x)}\right)}{h} \quad [\text{Using (i)}]$$

$$= \lim_{h \rightarrow 0} \left[\frac{f\left(\frac{h}{1+x(x+h)}\right)}{h} \right]$$

$$= \lim_{h \rightarrow 0} \frac{f\left(\frac{h}{1+xh+x^2}\right)}{\left(\frac{h}{1+xh+x^2}\right)} \times \left(\frac{1}{1+xh+x^2}\right)$$

$$= \lim_{h \rightarrow 0} \frac{f\left(\frac{h}{1+xh+x^2}\right)}{\left(\frac{h}{1+xh+x^2}\right)} \times \lim_{h \rightarrow 0} \left(\frac{1}{1+xh+x^2}\right)$$

$$\left(\text{Using } \lim_{x \rightarrow 0} \frac{f(x)}{x} = 2\right)$$

$$= 2 \times \frac{1}{1+x^2} = \frac{2}{1+x^2}$$

Integration both sides, we get

$$f(x) = 2\tan^{-1}(x) + c, \text{ where } f(0) = 0 \Rightarrow c = 0$$

Thus,

$$f\left(\frac{1}{\sqrt{3}}\right) = 2\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = 2 \cdot \frac{\pi}{6} = \frac{\pi}{3}$$

$$\text{And, } f'(1) = \frac{2}{1+1^2} = \frac{2}{2} = 1$$

9. Ans (6)

$$OP = OM$$

$$y - x \frac{dy}{dx} = \sqrt{x^2 + y^2}$$

$$\frac{dy}{dx} = \frac{y - \sqrt{x^2 + y^2}}{x}$$

$$\frac{dy}{dx} = \frac{y}{x} - \sqrt{1 + \left(\frac{y}{x}\right)^2}$$

$$\text{Putting } \frac{y}{x} = v \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\therefore v + x \frac{dv}{dx} = v - \sqrt{1 + v^2}$$

$$\Rightarrow \log(v + \sqrt{1 + v^2}) = \log \frac{c}{x}$$

$$\Rightarrow (v + \sqrt{1 + v^2}) = \frac{c}{x}$$

$$\Rightarrow y + \sqrt{x^2 + y^2} = c$$

Curve is parabola so $e = 1$

10. Ans (4)

$$x^2 \frac{dy}{dx} - xy = 1 + \cos\left(\frac{y}{x}\right)$$

$$\text{or } \frac{x(xdy - ydx)}{dx} = 1 + \cos\left(\frac{y}{x}\right)$$

$$\Rightarrow \frac{(xdy - ydx)}{x^2} = \frac{dx}{x^3}$$

$$\Rightarrow \int \frac{d\left(\frac{y}{x}\right)}{1 + \cos\left(\frac{y}{x}\right)} = \int \frac{dx}{x^3}$$

$$\Rightarrow \frac{1}{2} \int \frac{d\left(\frac{y}{x}\right)}{\cos^2\left(\frac{y}{2x}\right)} = \int \frac{dx}{x^3}$$

$$\Rightarrow \frac{1}{2} \frac{\tan\left(\frac{y}{2x}\right)}{\frac{1}{2}} = \frac{x^{-2}}{-2} + c$$

$$\Rightarrow \tan\left(\frac{y}{2x}\right) + \frac{1}{2x^2} = c$$

11. Ans (7)

We have

$$a^2 - (2x^2 + 1)a + x^4 + x = 0$$

$\therefore$

$$a = \frac{(2x^2 + 1) \pm \sqrt{(2x^2 + 1)^2 - 4(x^4 + x)}}{2}$$

$$\therefore 2a = (2x^2 + 1) \pm (2x - 1)$$

positive sign $a = x^2 + x$

negative sign $2a = 2x^2 - 2x + 2$

$$a = x^2 - x + 1$$

$$\text{if } x^2 + x - a = 0 \Rightarrow x = \frac{-1 \pm \sqrt{1 + 4a}}{2}$$

if $x^2 - x + 1 - a$

$$\Rightarrow x = \frac{1 \pm \sqrt{1 - 4 + 4a}}{2} = \frac{1 \pm \sqrt{4a - 3}}{2}$$

for x to be real $a \geq 3/4$ and $a \geq -1/4$

$$\Rightarrow a \geq 3/4 \Rightarrow 3 + 4 = 7$$

12. Ans (2)

$$P(E) = \frac{1}{2} \quad P(F_k) = {}^nC_k \times \frac{1}{2^n}$$

$$P(E \cap F_k) = {}^{n-1}C_{k-1} \times \left(\frac{1}{2}\right)^n$$

$$P(E \cap F_k) = P(E) \times P(F_k)$$

$${}^{n-1}C_{k-1} \times \left(\frac{1}{2}\right)^n = \left(\frac{1}{2}\right)^n \times \frac{1}{2} \times {}^nC_k$$

$$\text{or } 2 \times {}^{n-1}C_{k-1} = {}^nC_k$$

$$\frac{{}^nC_k}{{}^{n-1}C_{k-1}} = 2 = \frac{n}{k}$$

13. Ans (4)

Required no. of ways

$$= \text{coeff of } x^{30} \text{ in } (x^2 + x^3 + \dots + x^8)^5$$

$$= \text{coeff of } x^{30} \text{ in } \left(\frac{x^2(1-x^7)}{1-x}\right)^5$$

$$= \text{coeff of } x^{20} \text{ in } (1-x^7)^5(1-x)^{-5}$$

$$= {}^{24}C_{20} - 5 \times {}^{17}C_{13} + 10 \times {}^{10}C_6 = 826$$

$$8 + 2 - 6 = 4$$

14. **Ans (4)**

Given $\alpha + \beta = 1$... (i)

$2\alpha^2 + 2\beta^2 = 1$... (ii)

Solving equation (i) and (ii), we get

$$\alpha = \beta = \frac{1}{2}$$

$$Q = \frac{\alpha}{\beta} = 1 \quad \dots \text{(iii)}$$

And given

$$f(2+x) + f(x) = 2 \quad \forall x \in [0, 2] \quad \dots \text{(iv)}$$

$$\text{Now, } p = \int_0^4 f(x) dx - 4$$

$$= \int_0^2 f(x) dx + \int_2^4 f(x) dx - 4$$

$$= \int_0^2 f(x) dx + \int_0^2 f(t+2) dt - 4$$

{Let, $x = t + 2$ for second integration}

$$= \int_0^2 f(x) dx + \int_0^2 \{2 - f(x)\} dx - 4$$

$$= \int_0^2 f(x) dx + 2 \int_0^2 dx - \int_0^2 f(x) dx - 4 = 0$$

Then, $p = 0, q = 1$

Let the roots of equation $ax^2 - bx + c = 0$ be α and β

$$\therefore f(x) = ax^2 - bx + c = a(x - \alpha)(x - \beta) \quad \dots \text{(v)}$$

Since, equation $f(x) = 0$ has both roots between 0 and 1

$$\therefore f(0).f(1) > 0 \quad \dots \text{(vi)}$$

$$\text{But } f(0).f(1) = c(a - b + c) = \text{an integer} \quad \dots \text{(vii)}$$

$$\therefore \text{least value of } f(0).f(1) = 1 \quad \dots \text{(viii)}$$

Now, from equation (v)

$$F(0).f(1) = a\alpha\beta a(1 - \alpha)(1 - \beta)$$

$$= a^2\alpha\beta(1 - \alpha)(1 - \beta) \quad \dots \text{(ix)}$$

As we know,

$$a(1 - \alpha) \text{ as greatest value } \frac{1}{4} \text{ at } \alpha = \frac{1}{2} \text{ and}$$

$$\beta(1 - \beta) \text{ has greatest value } \frac{1}{4} \text{ at } \beta = \frac{1}{2}.$$

But $\alpha \neq \beta$

Thus from (viii) greatest value of

$$f(0).f(1) < \frac{a^2}{16} \quad \dots \text{(x)}$$

$\therefore$ from (viii) and (x)

$$1 < \frac{a^2}{16}$$

$$\Rightarrow a^2 - 16 > 0$$

$$\Rightarrow a < -4 \text{ or } a > 4 \quad (\because a \in \mathbb{N})$$

$$\Rightarrow \text{least value of } a = 5 \text{ (as } a \in \text{Natural number)}$$